

From Texts to Networks

Worshop in Network Analysis

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Introduction

Aims of the study

- Assessing large number of texts
- Leveraging existing explanatory methods
- Automatic analysis of textual relations...
- ... that will reveal groups of similar texts

Mapping literature

- Chronology
- Genre
- Imitation
- Authorship
- ...

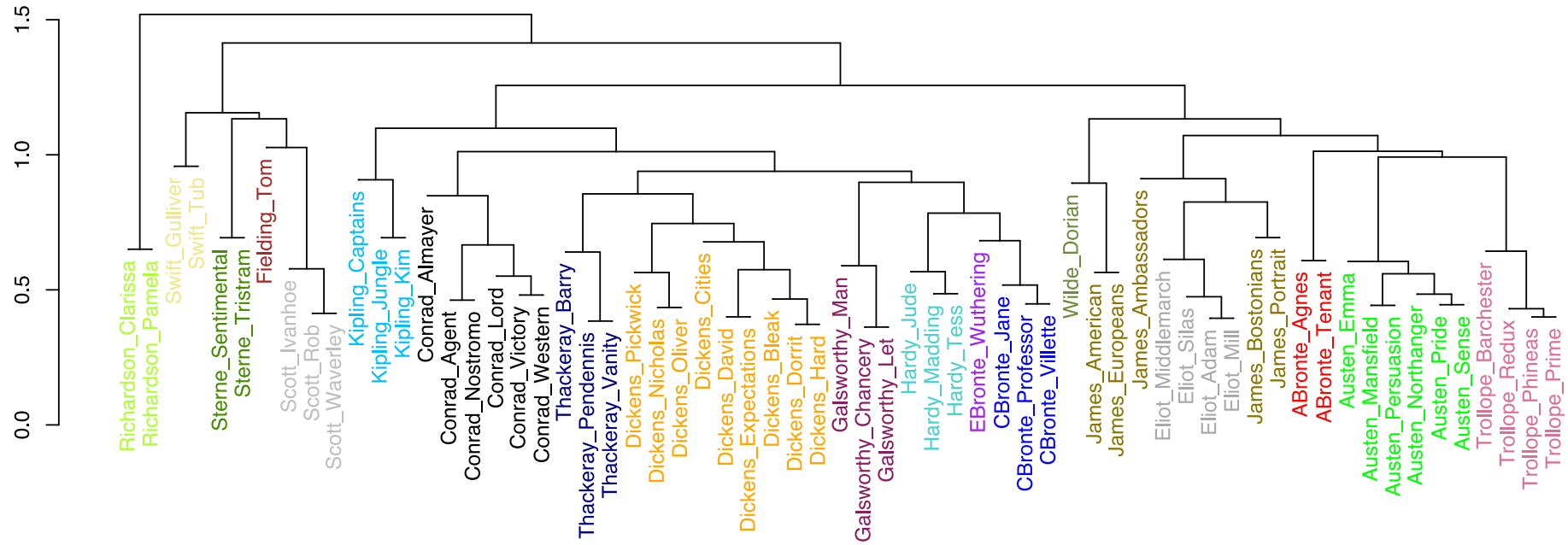
Basic concepts

- “Computation into criticism” (John Burrows)
- “Distant reading” (Franco Moretti)
- “Macroanalysis” (Matthew Jockers)
- Big Data
- Authorship attribution

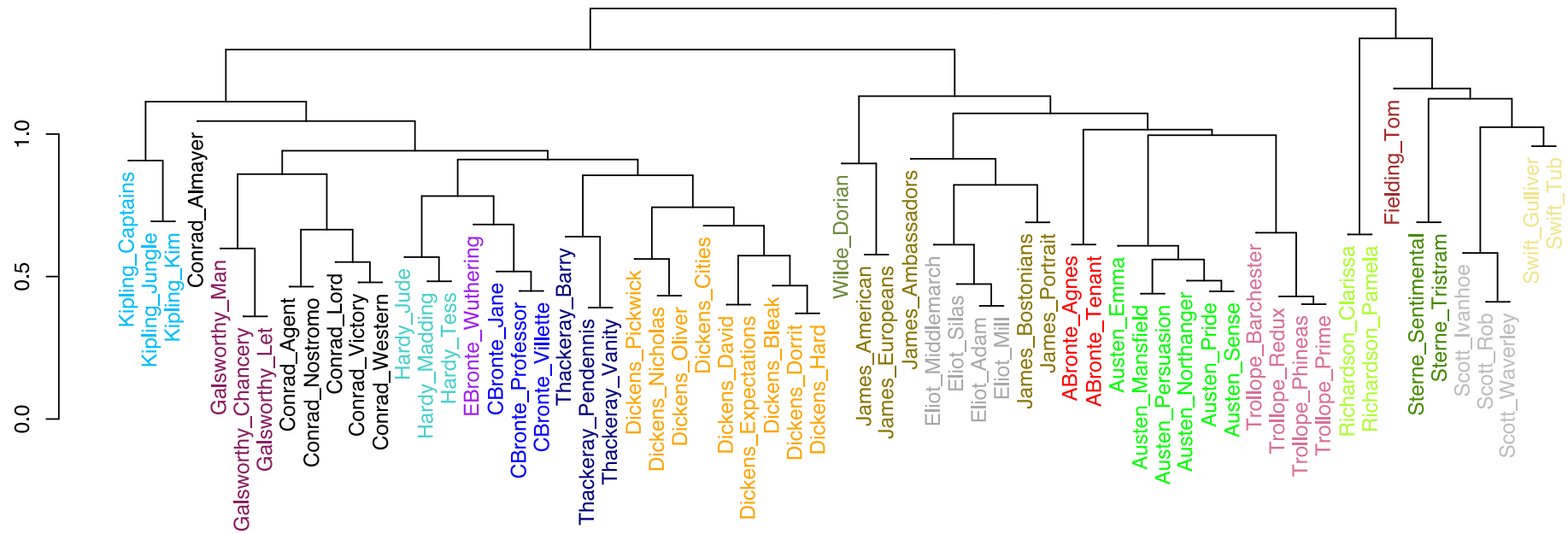
Cluster analysis dendrograms

- Highly dependent on feature selection
- Highly dependent on similarity measure
- Highly dependent on linkage algorithm
- No validation provided

136 most frequent words



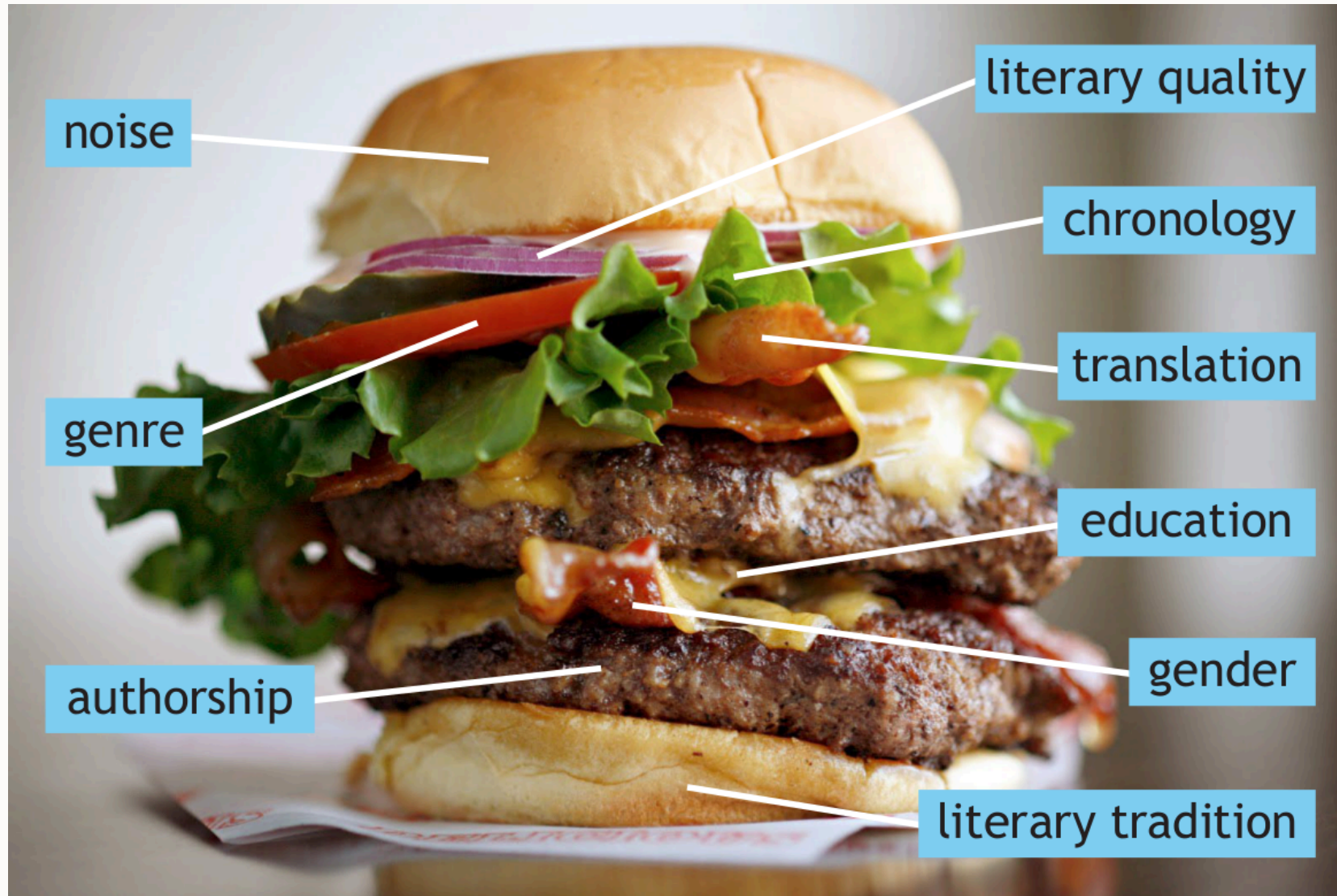
137 most frequent words



Unstable results

- Noise?
- Unreliability of the linkage procedure?
- More than one stylistic layer?

Layers



Networks to rescue

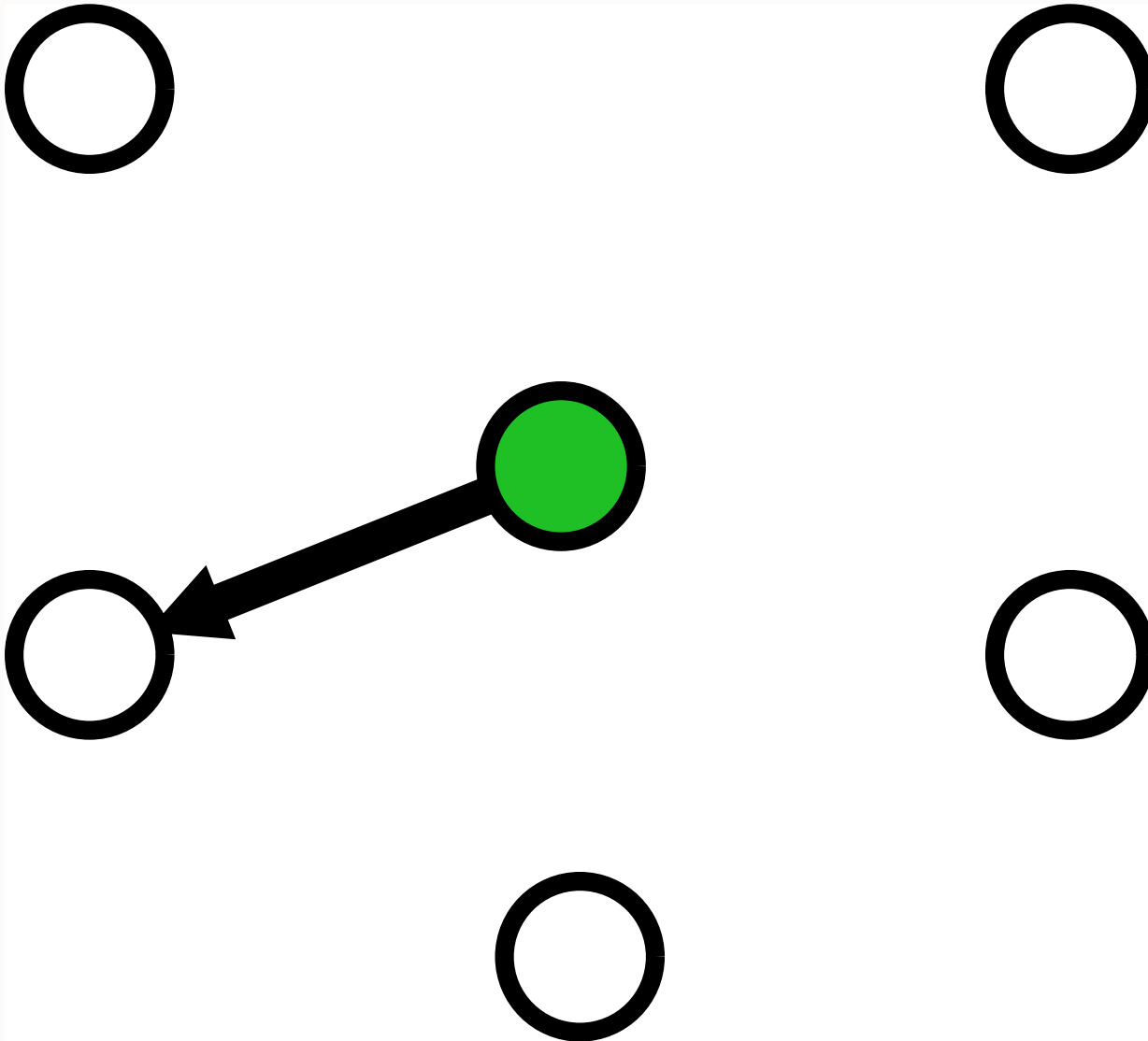
A new technique: assumptions

- Explanatory power of dendrograms
- Stable results and/or validation
- Scalability: 500 texts? 1000 texts? 31M texts?

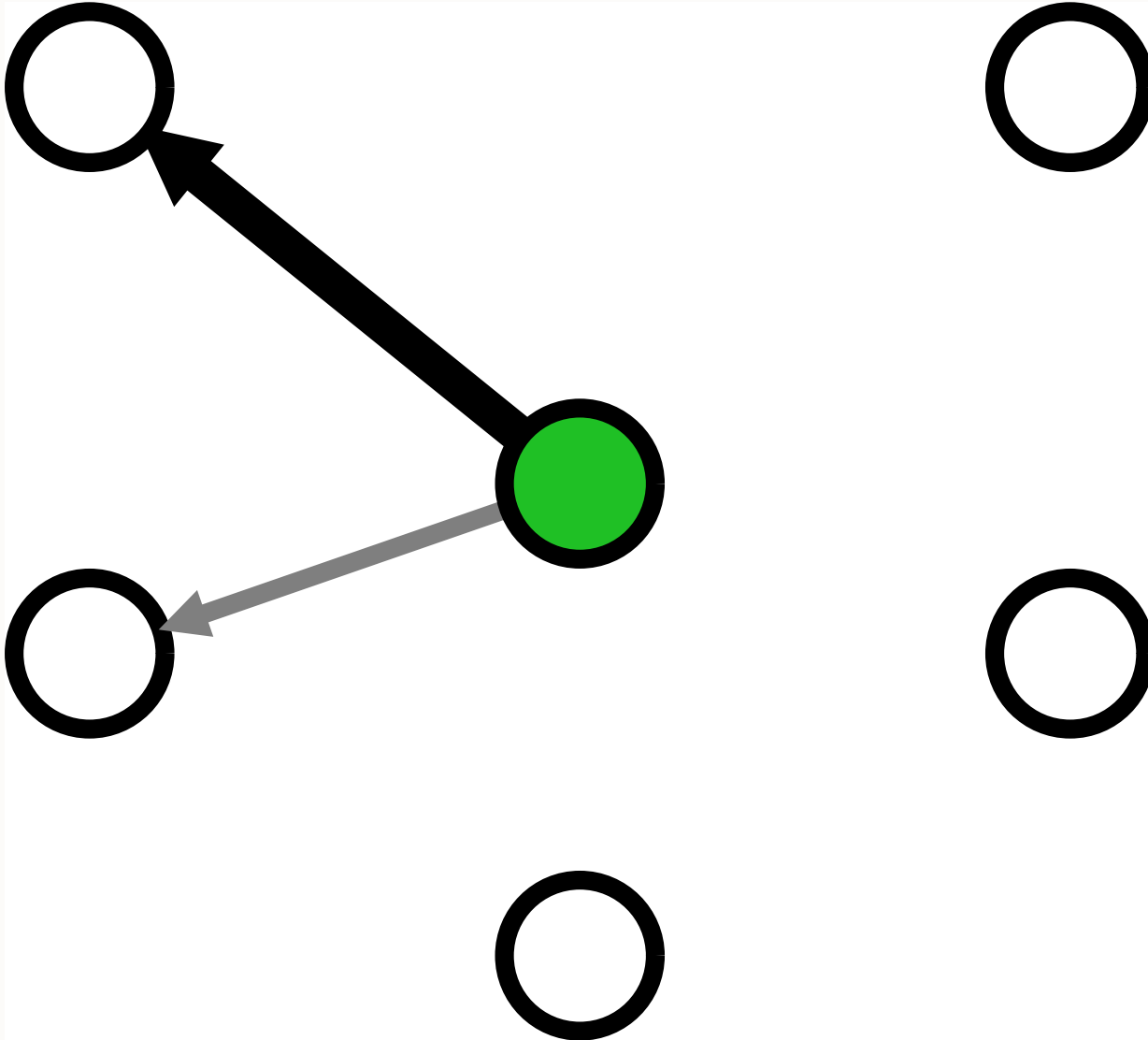
Authorship attribution

- An anonymous text is tested against a set of “candidates”
- Goal: to find the nearest neighbor
- Effective style-marker: most frequent words (grammatical words)

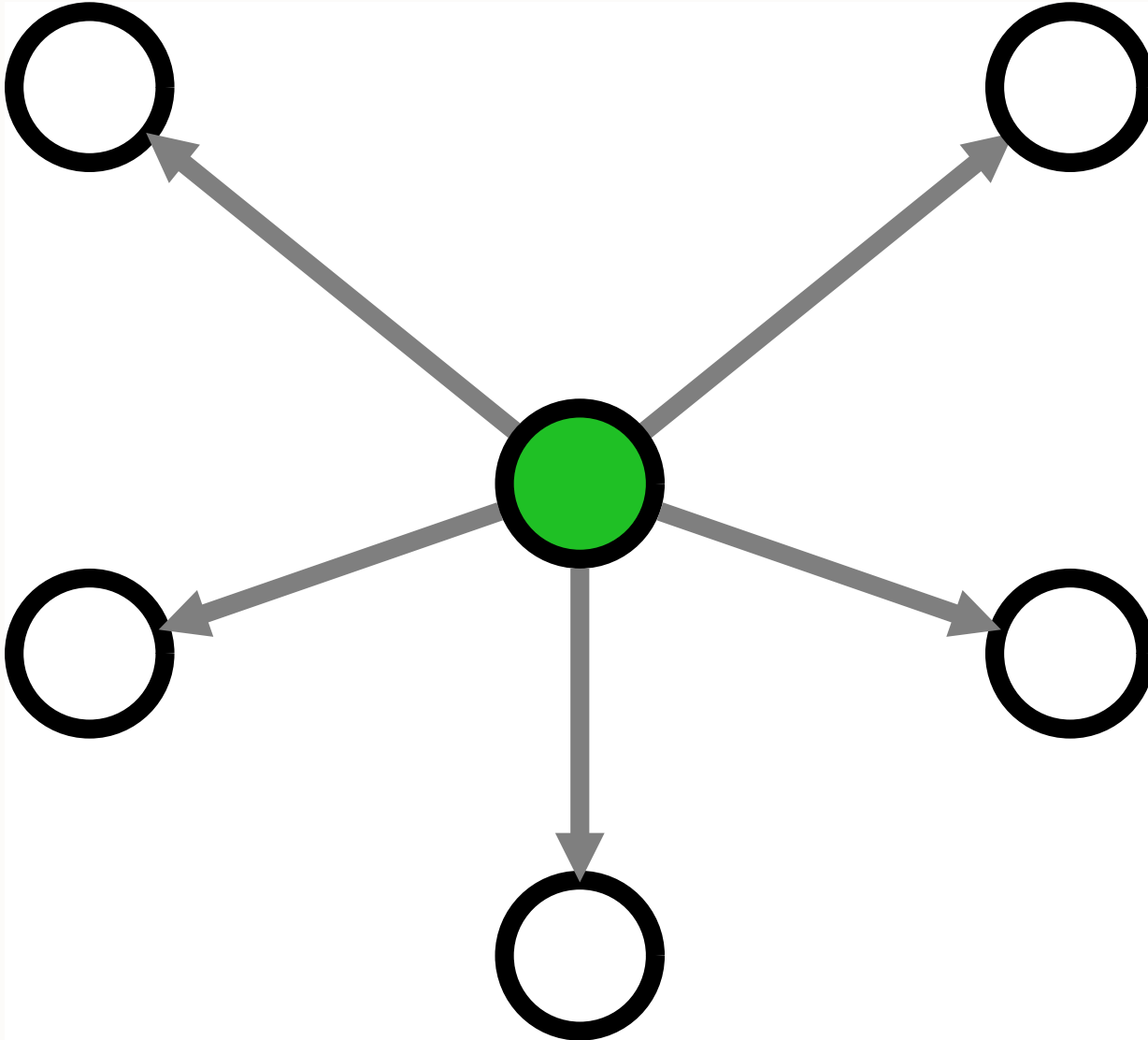
Establishing a distance



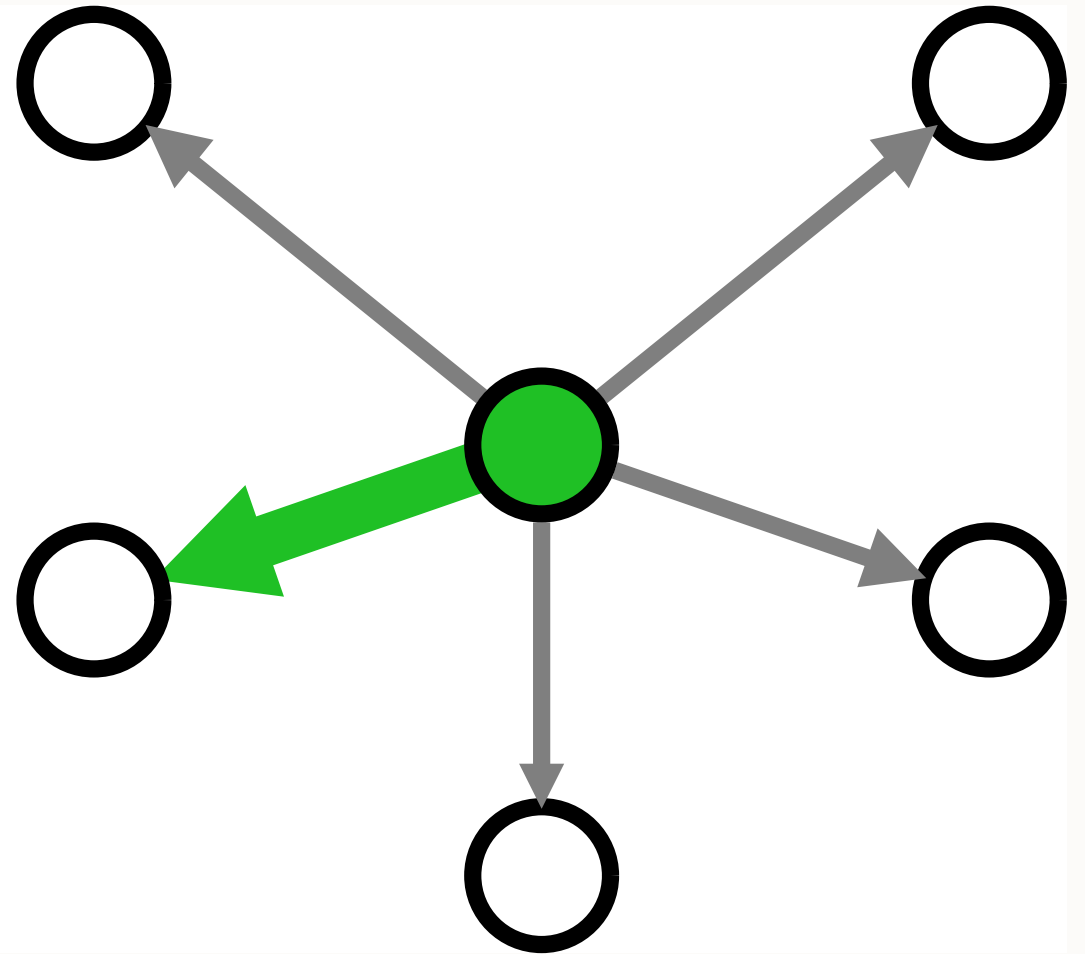
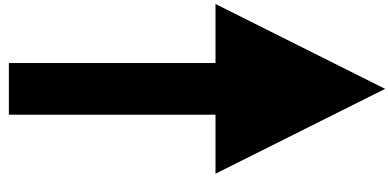
Another distance...



Checking each sample



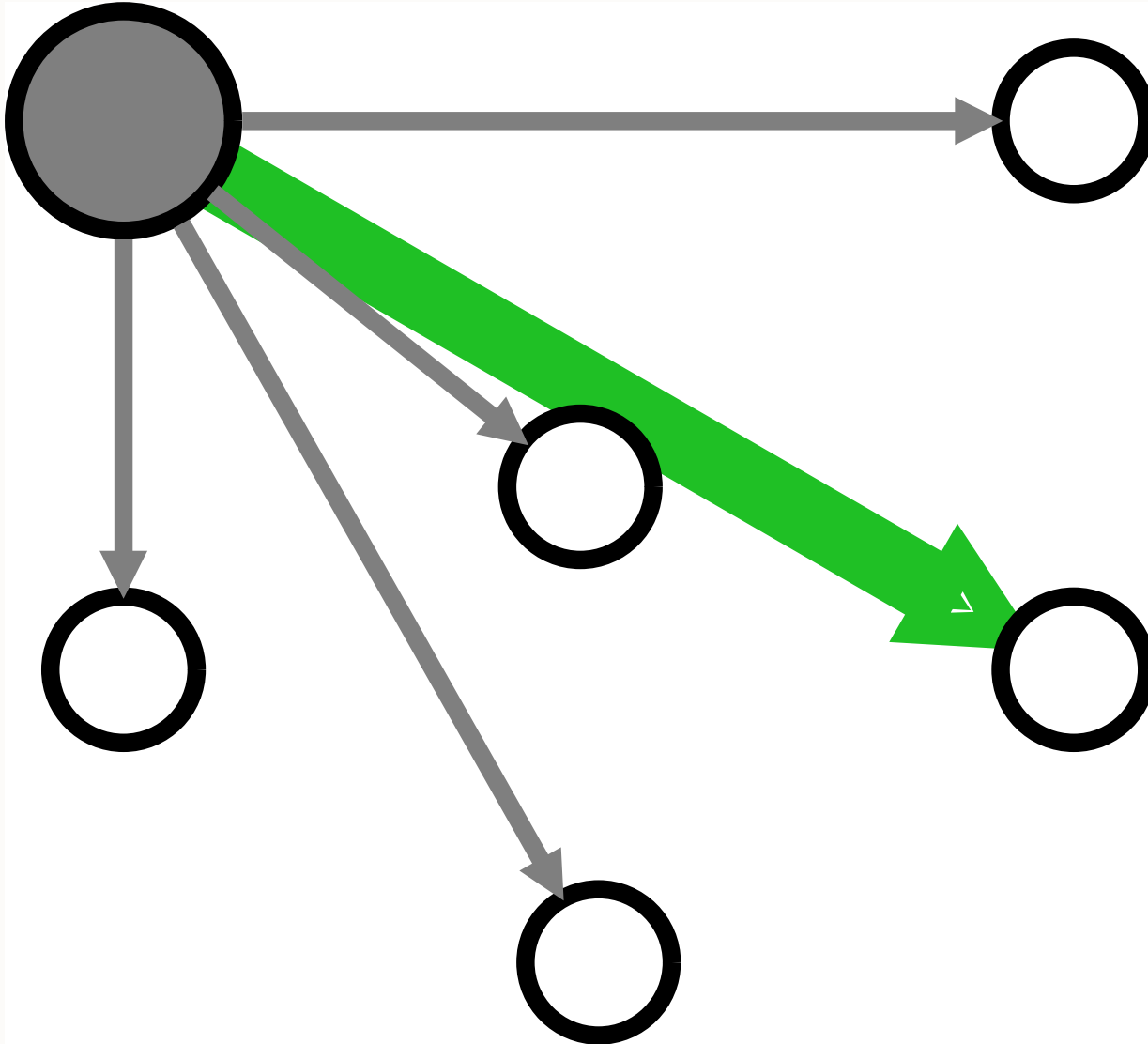
Deciding which distance is shortest



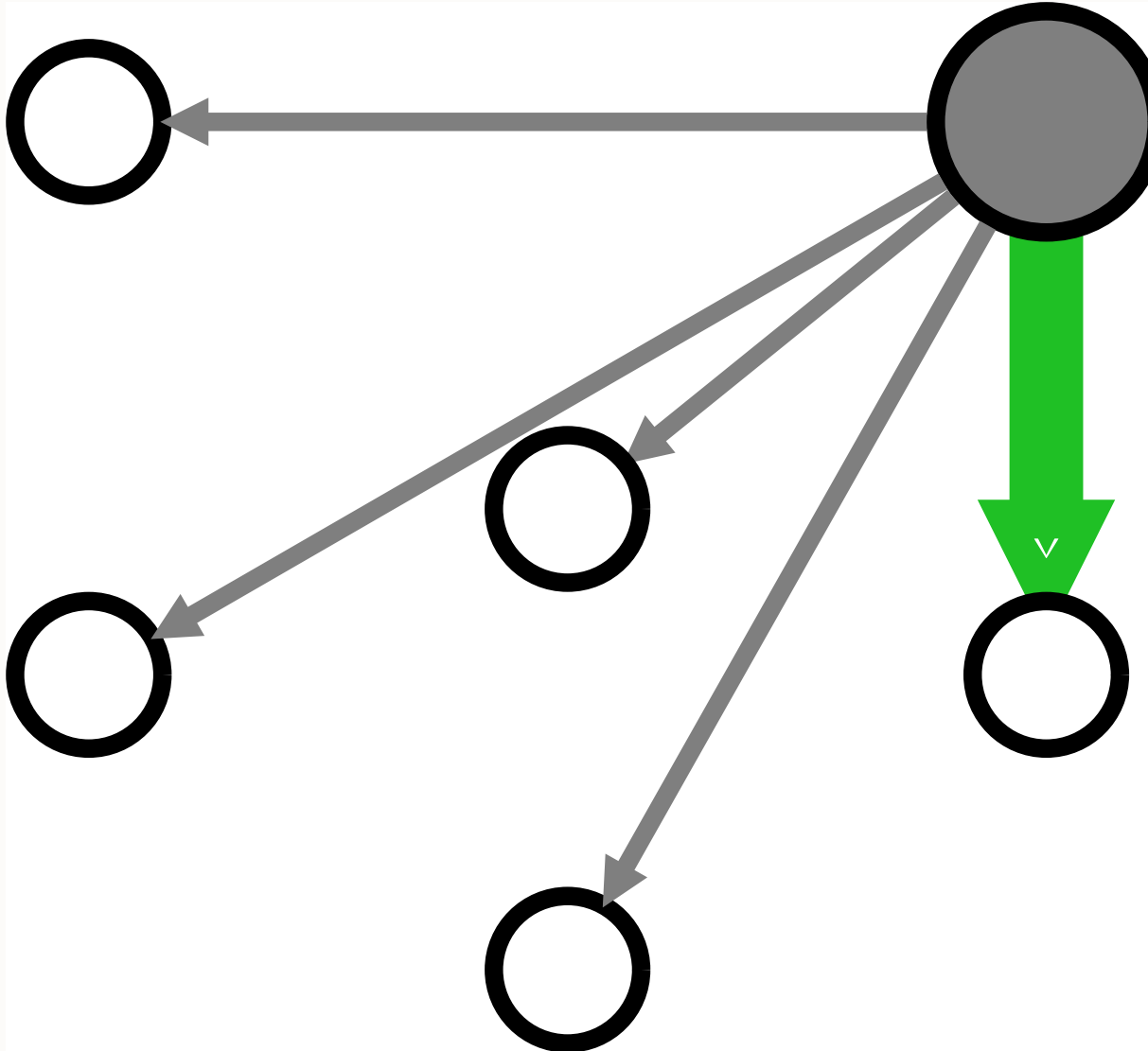
From attribution to stylometry

- If it works with anonymous texts, ...
- ... what about extending the same procedure?
- What about applying it to *all* texts?

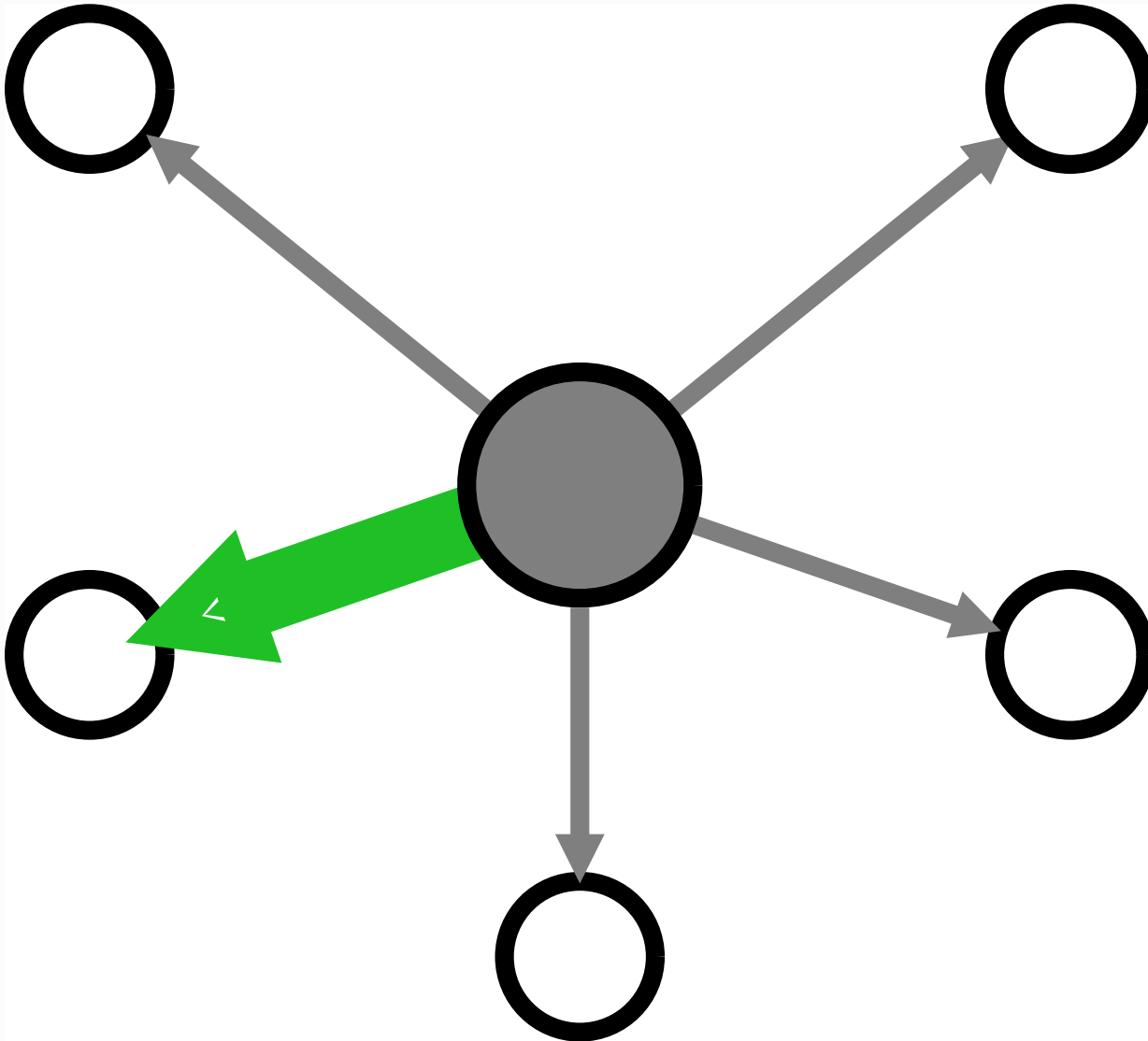
Distances to txt1



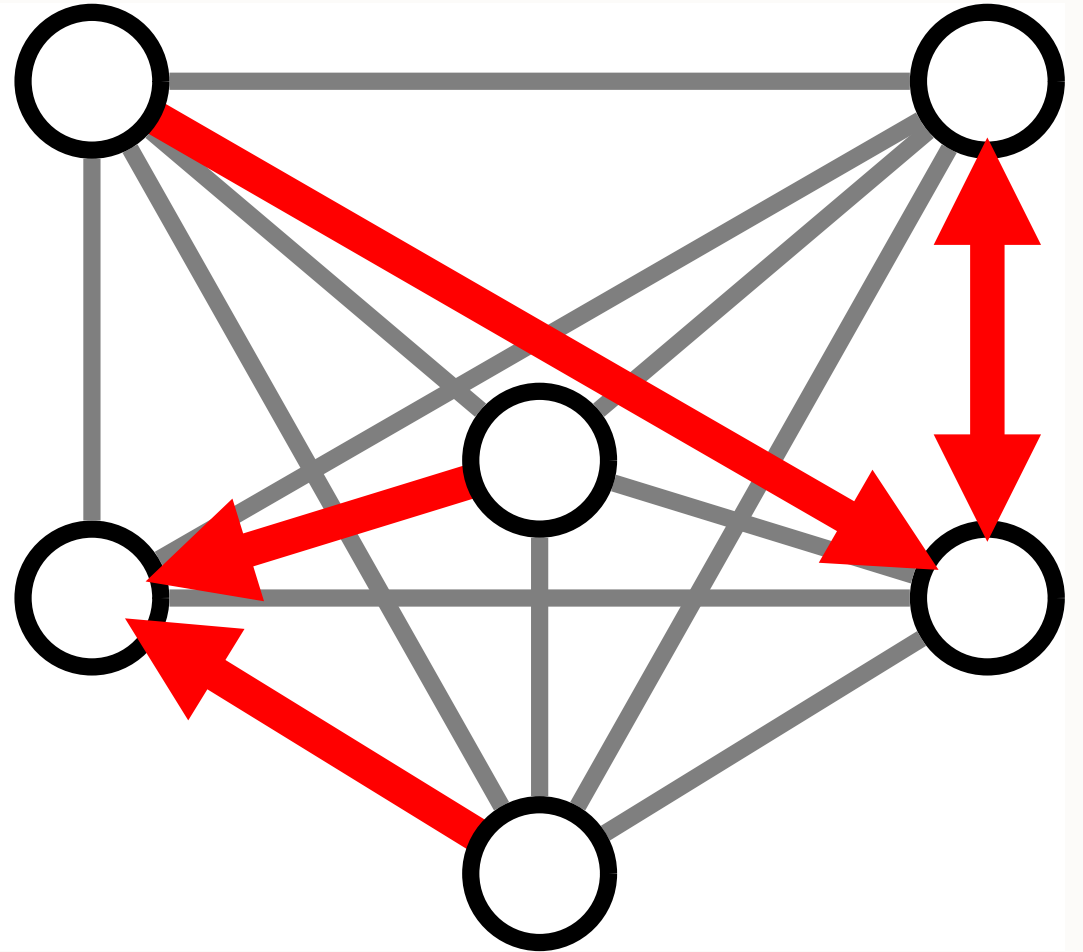
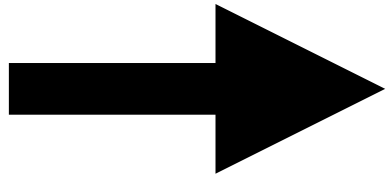
Distances to txt2



Distances to txt3



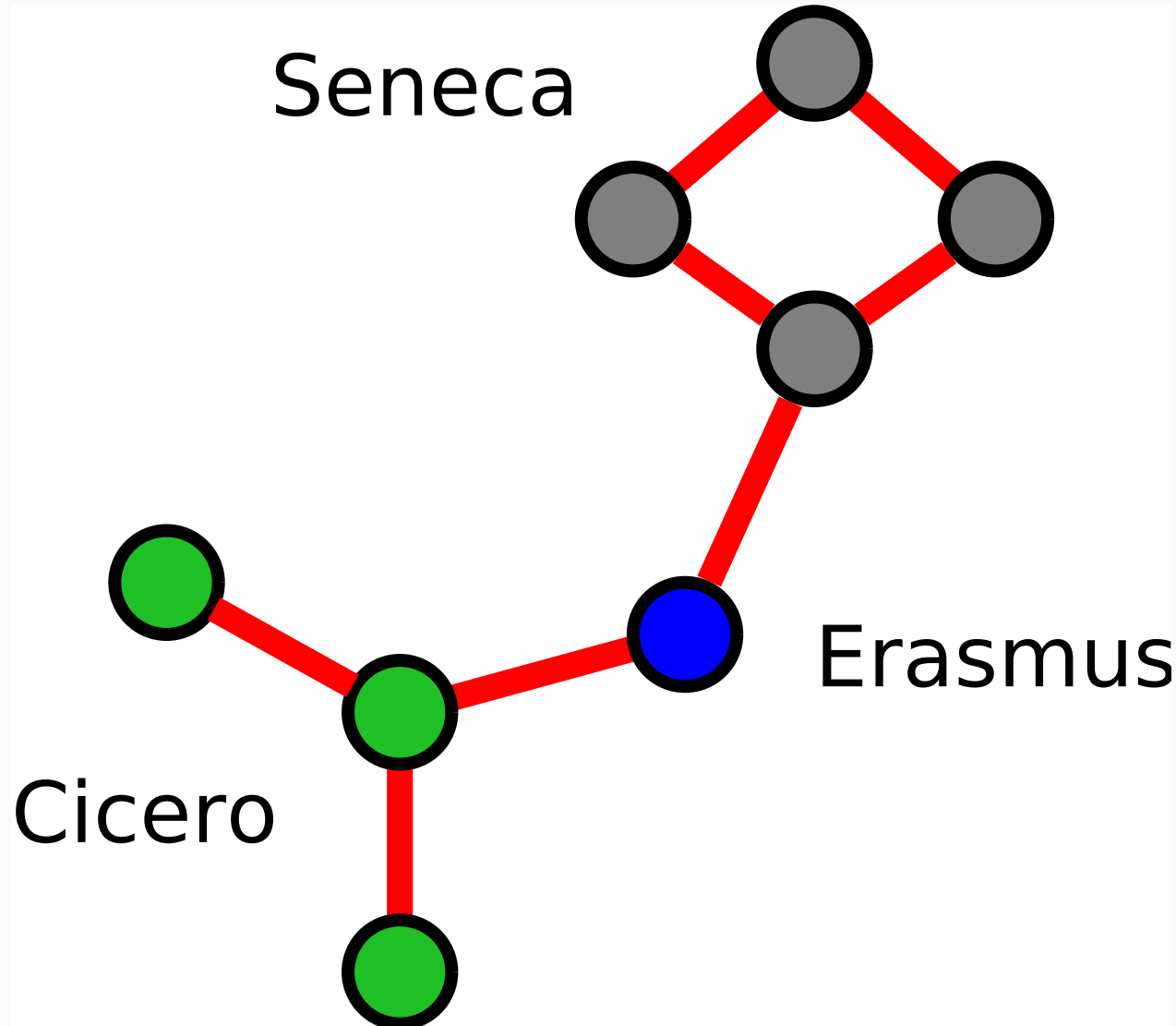
A resulting network



From stylometry to networks

- Nearest neighbors can be represented as connected nodes of a network.
- A variety of layout algorithms can be applied.

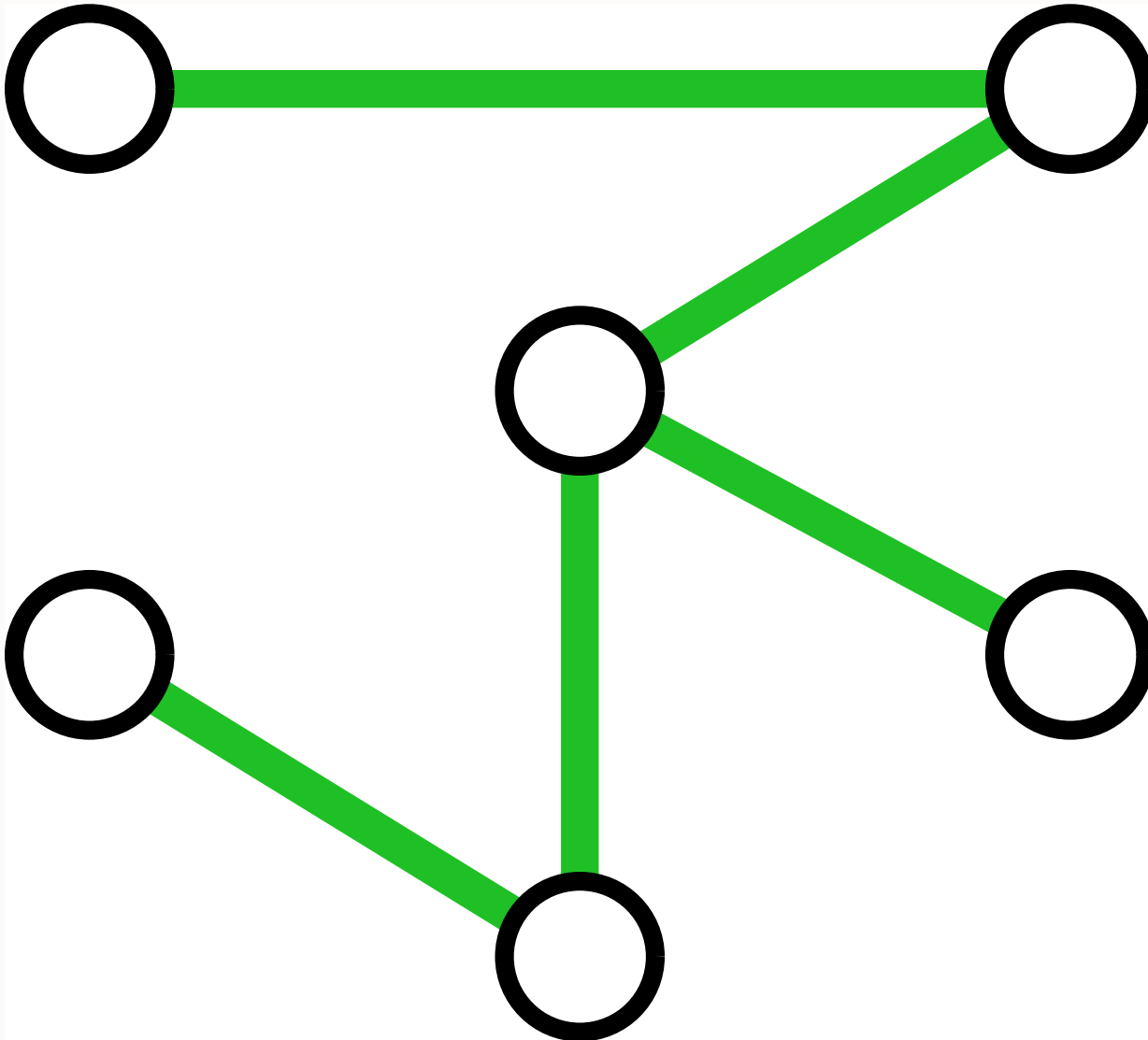
Example



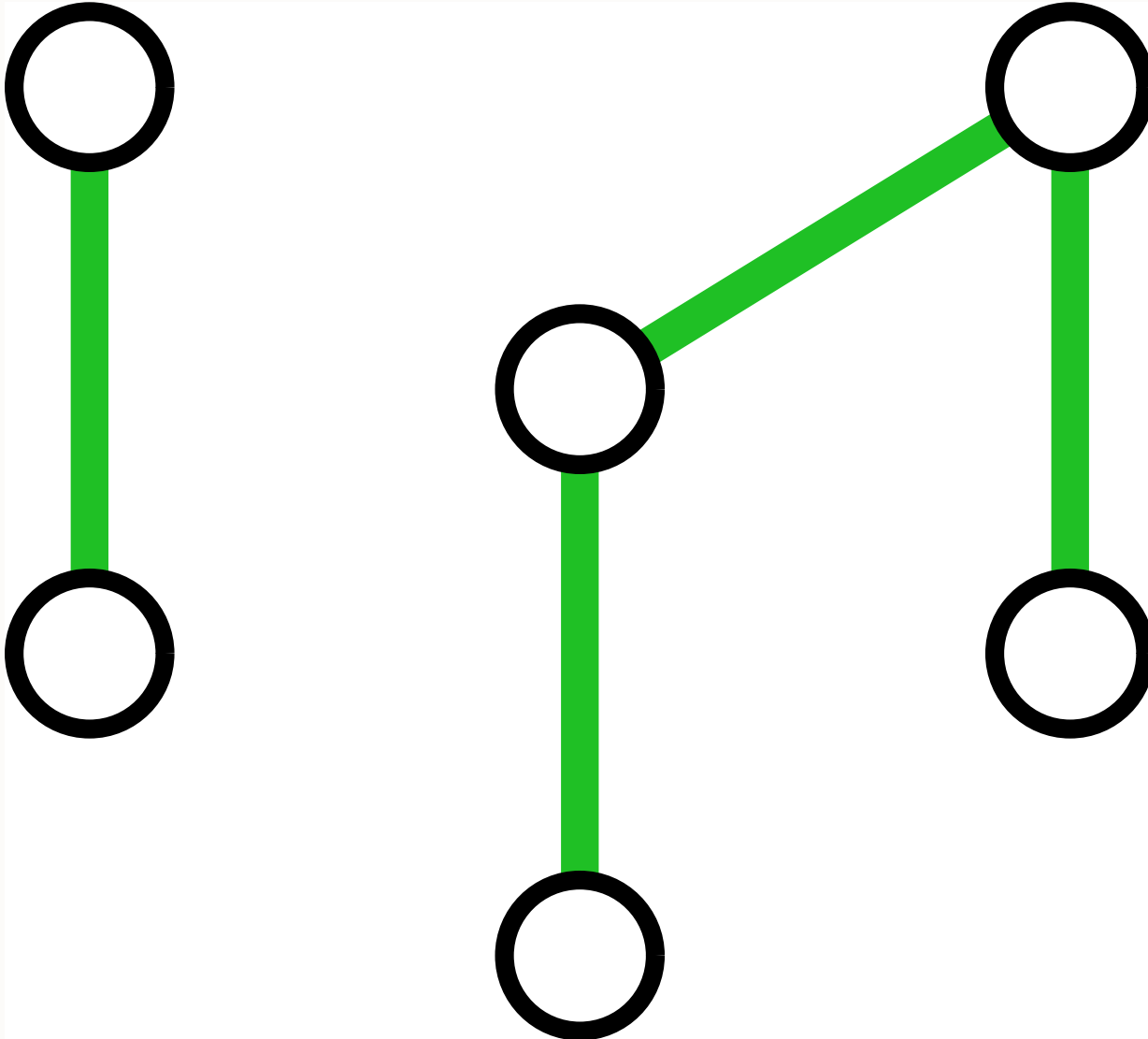
Reliability, stupid!

- Instead of one analysis (e.g. 100 MFWs)...
- ... a whole range (100, 200, 300 MFW, etc.).
- Next, particular “snapshots” summarized.

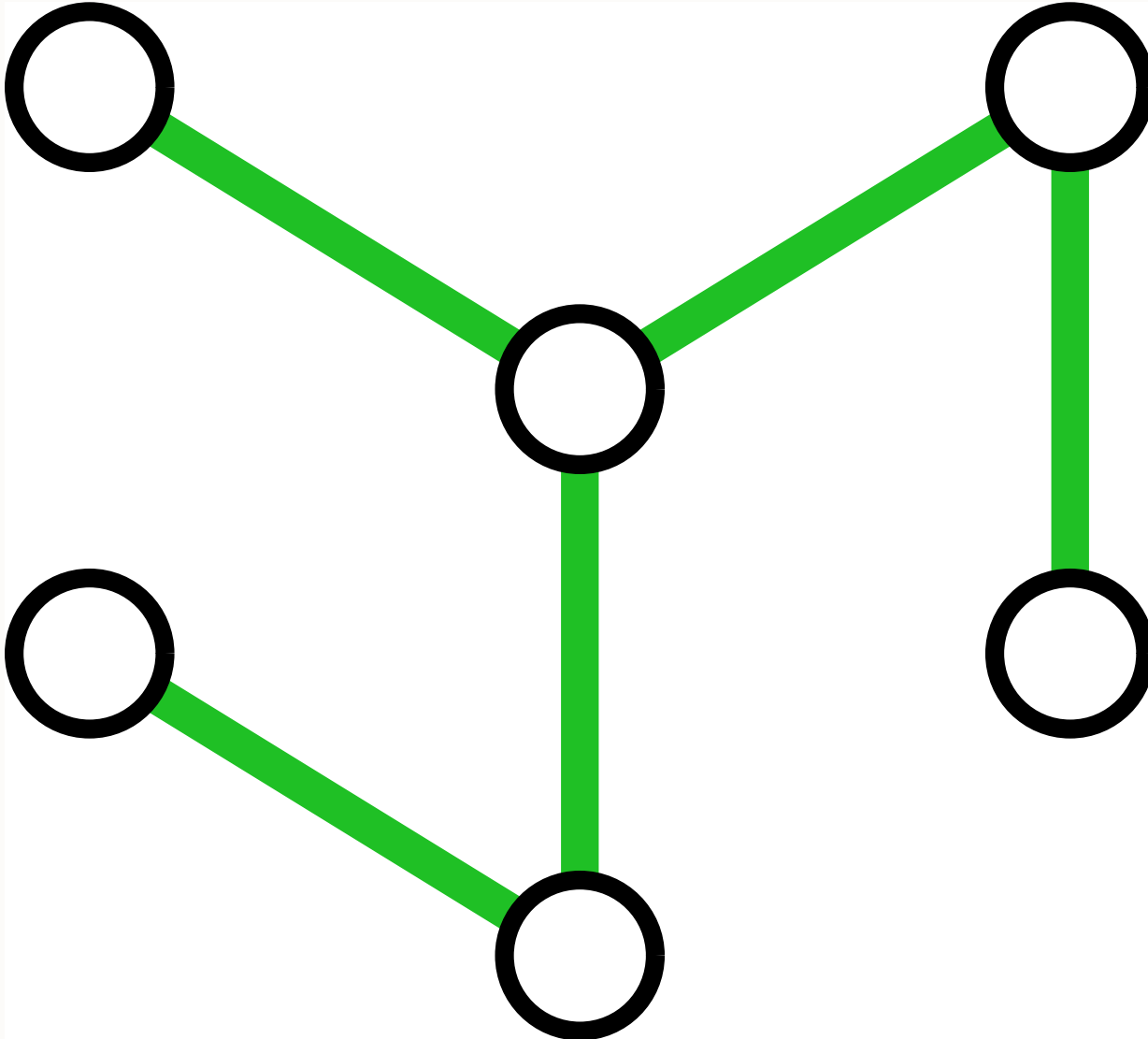
100 MFWs



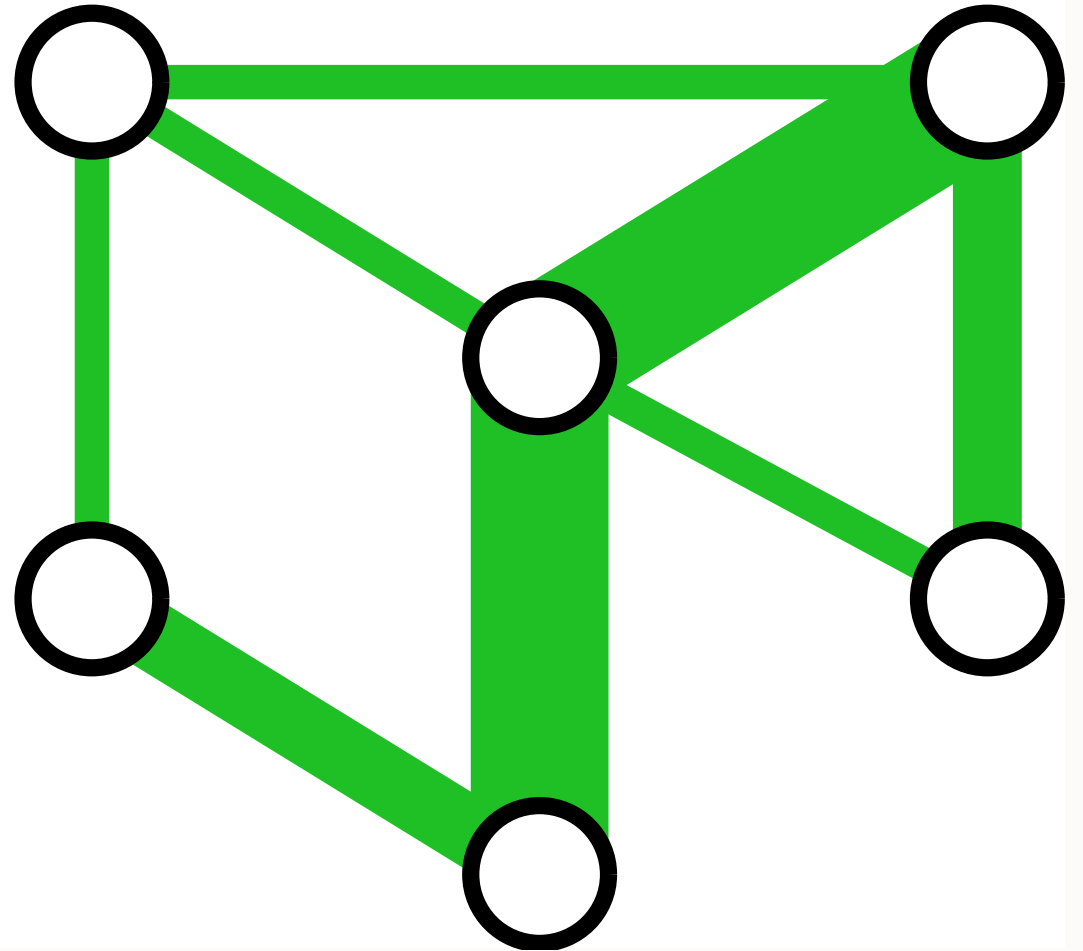
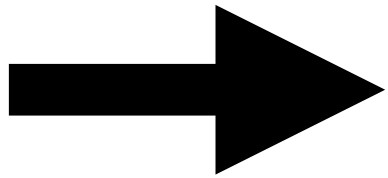
200 MFWs



300 MFWs



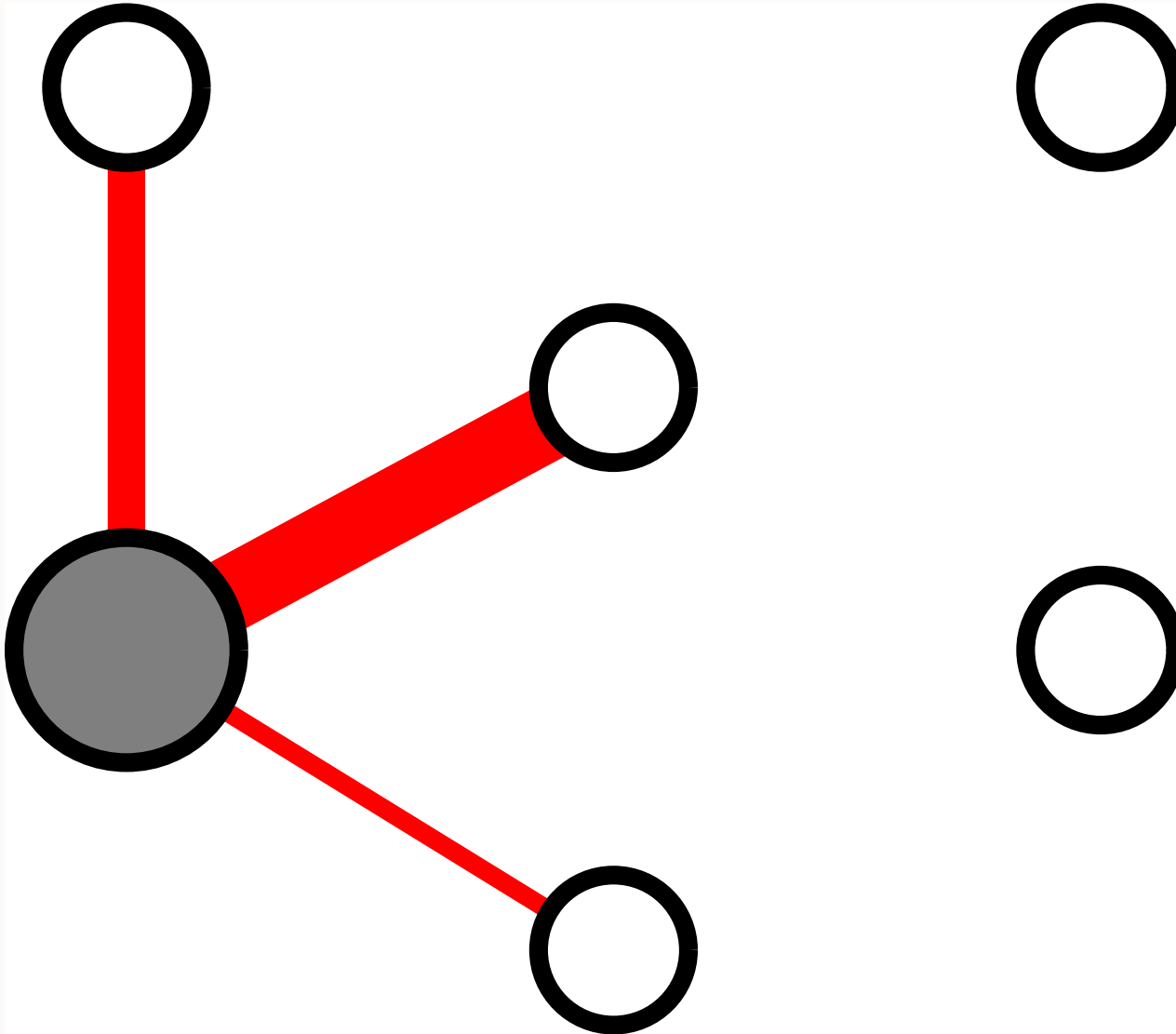
Consensus network



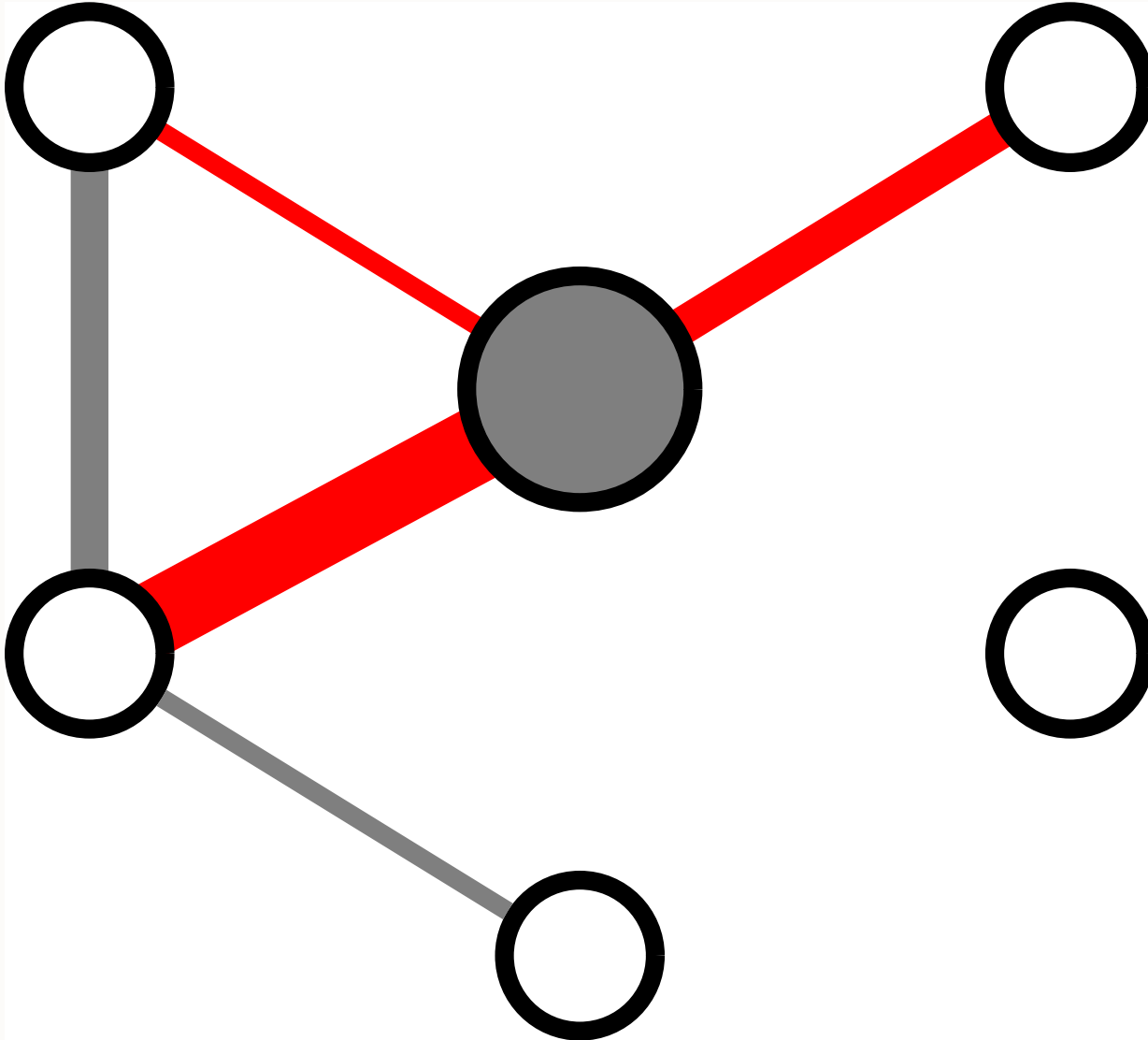
Importance of runners-up

- Nearest neighbors = the most similar
- Runners-up: do they really deserve to be filtered out?
- Solution: more connections!

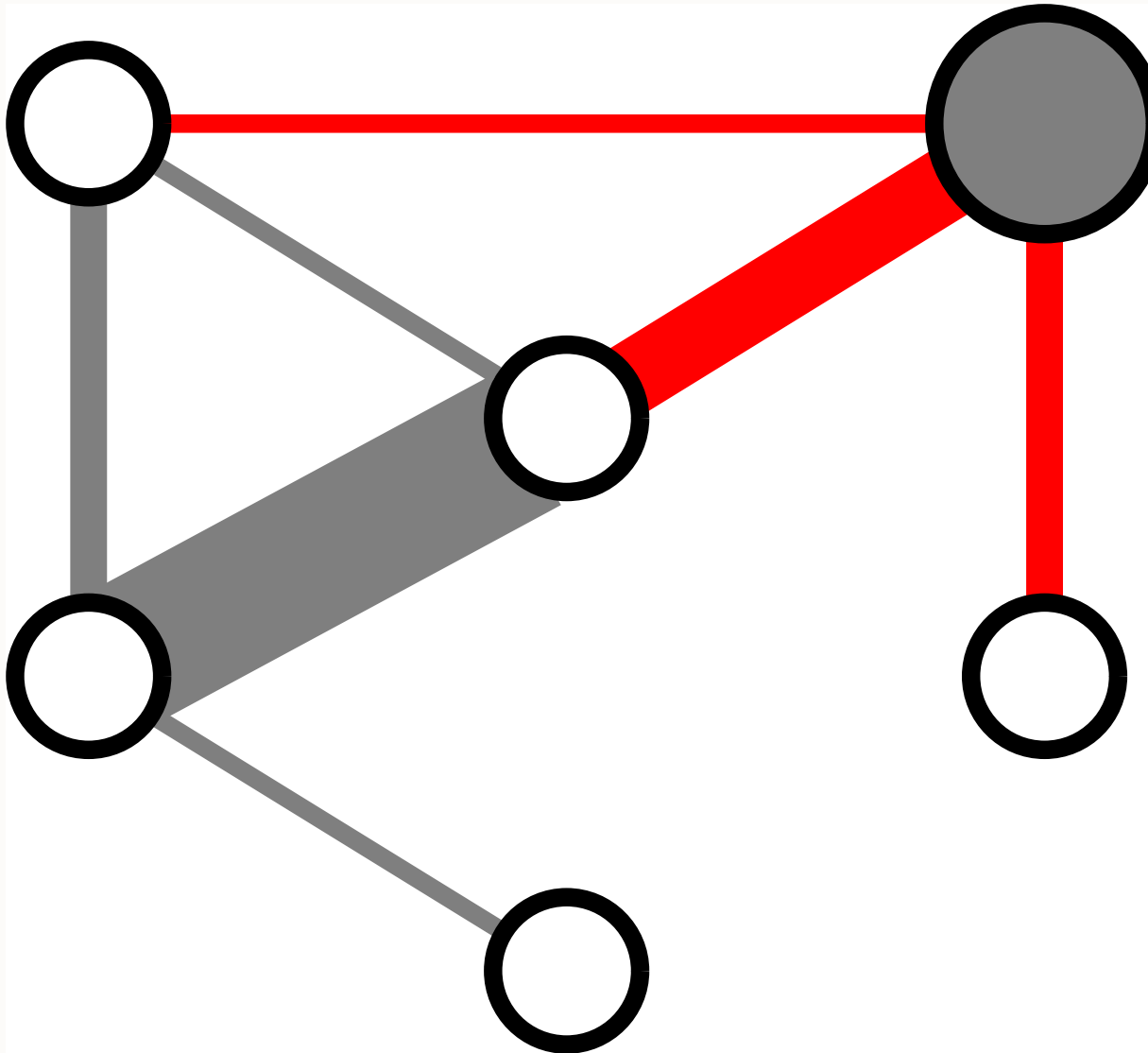
3 neighbors to txt1



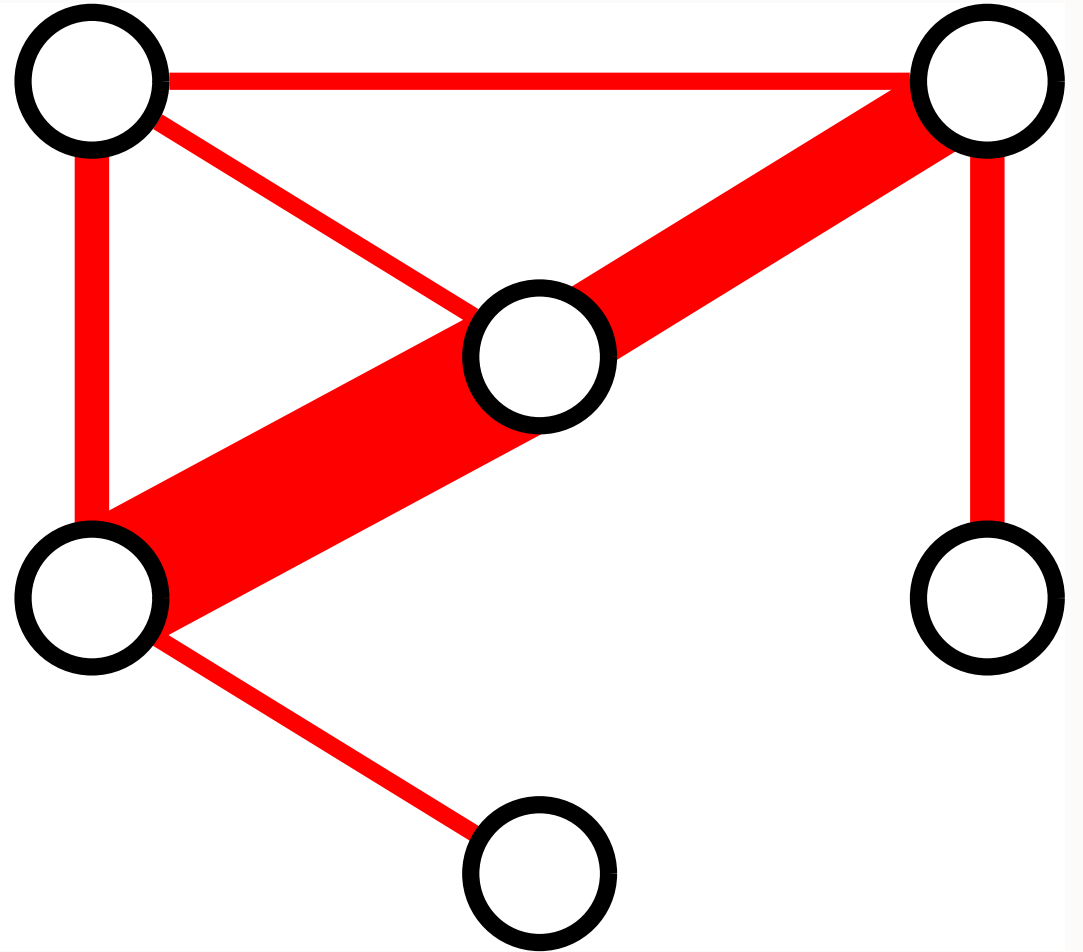
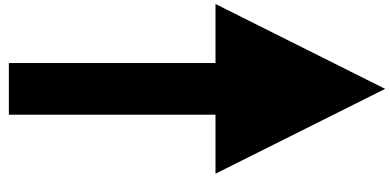
3 neighbors to txt2



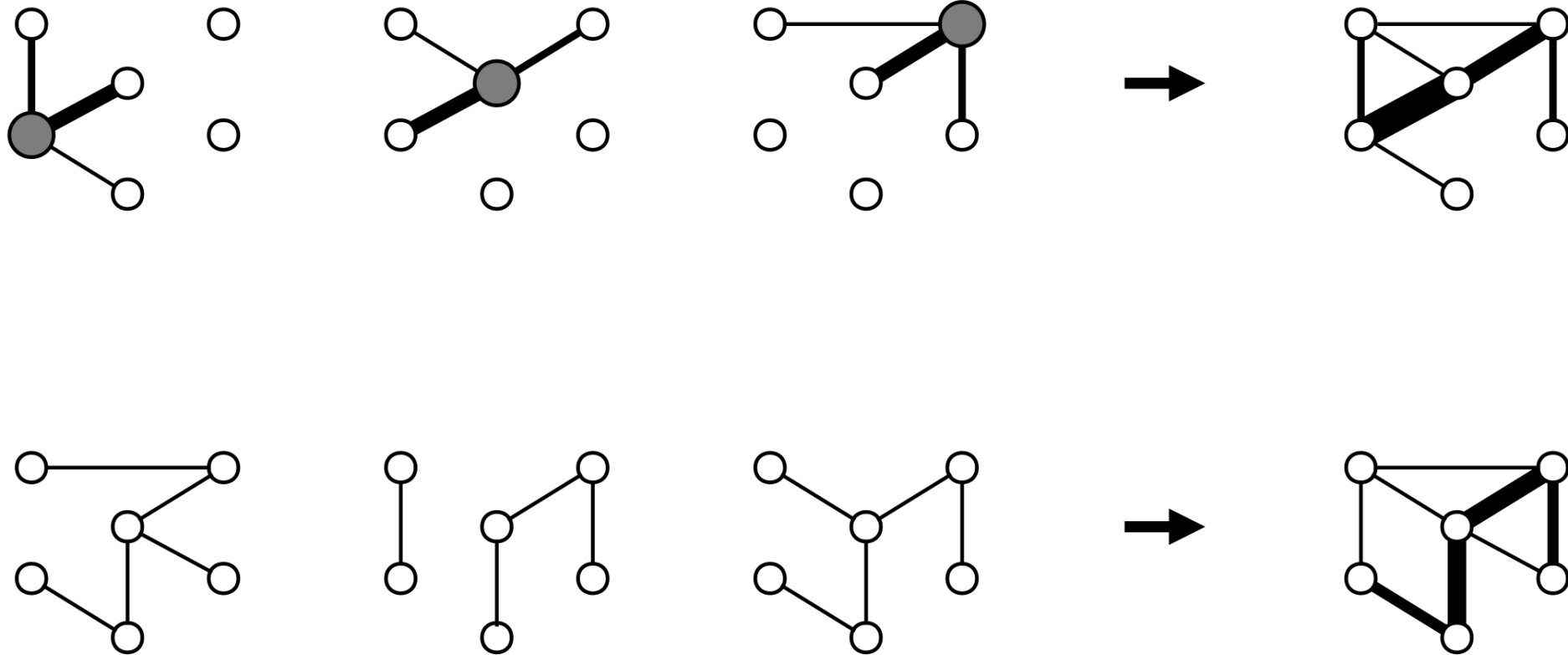
3 neighbors to txt3



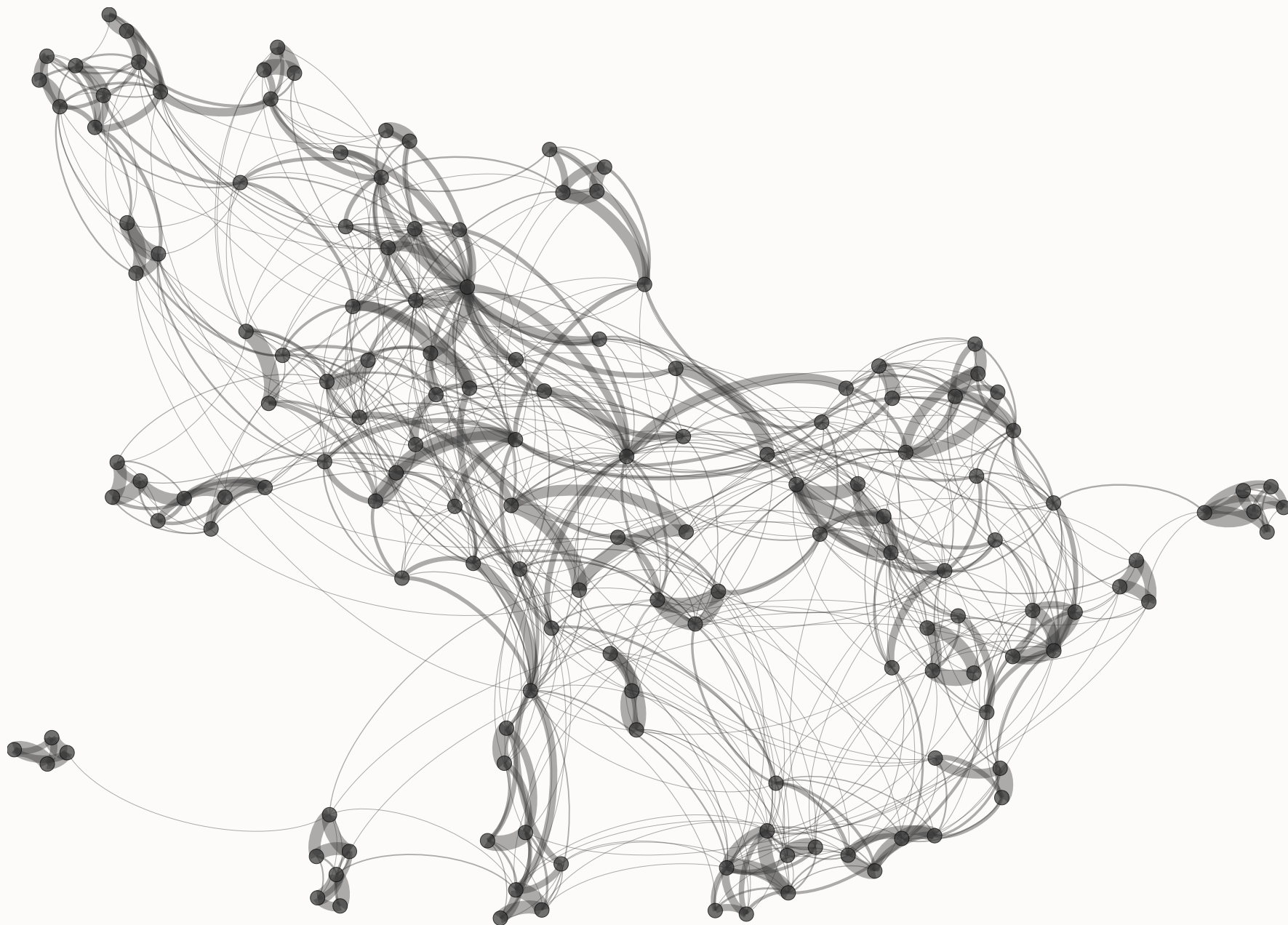
A resulting network



Two algorithms together

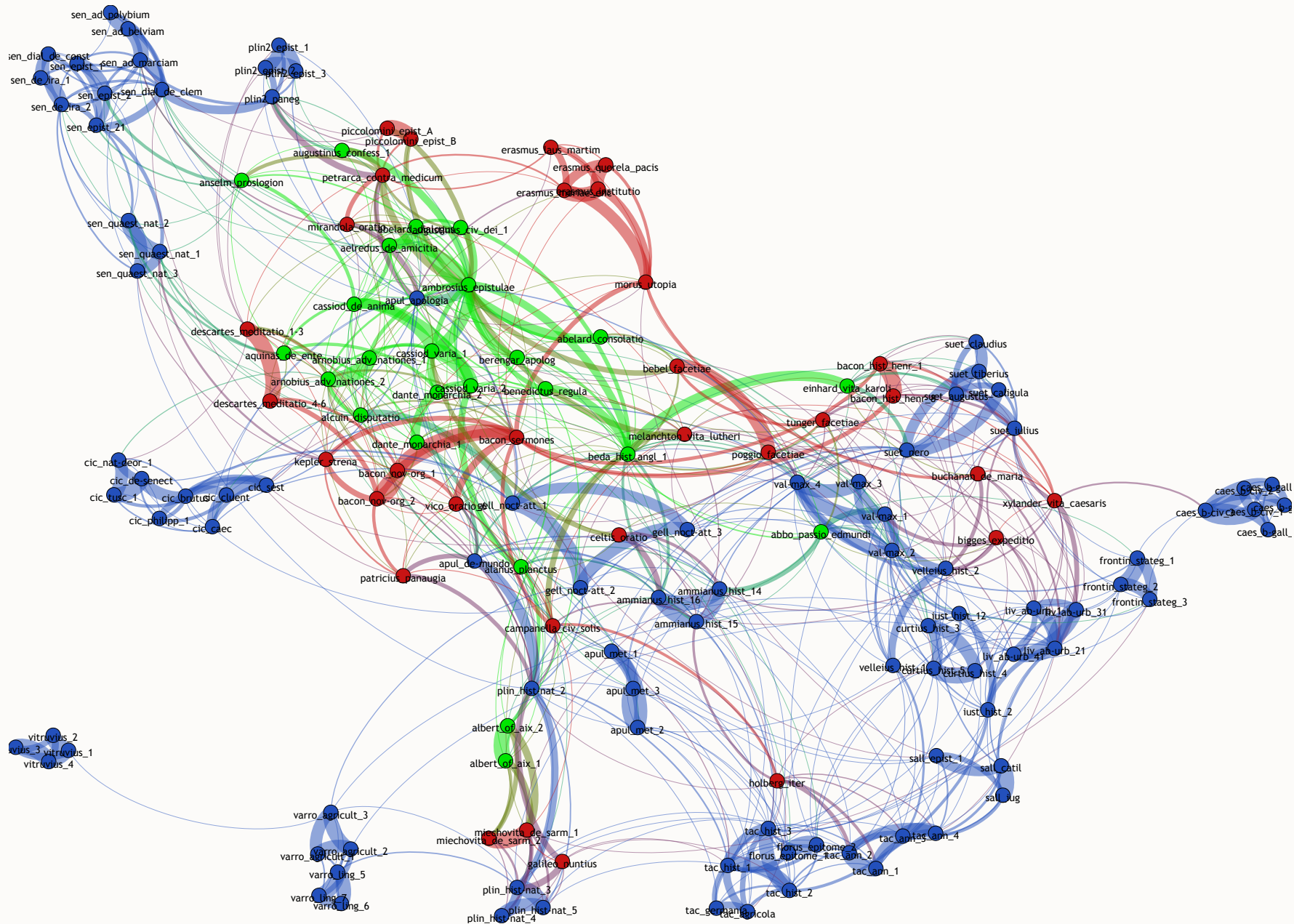


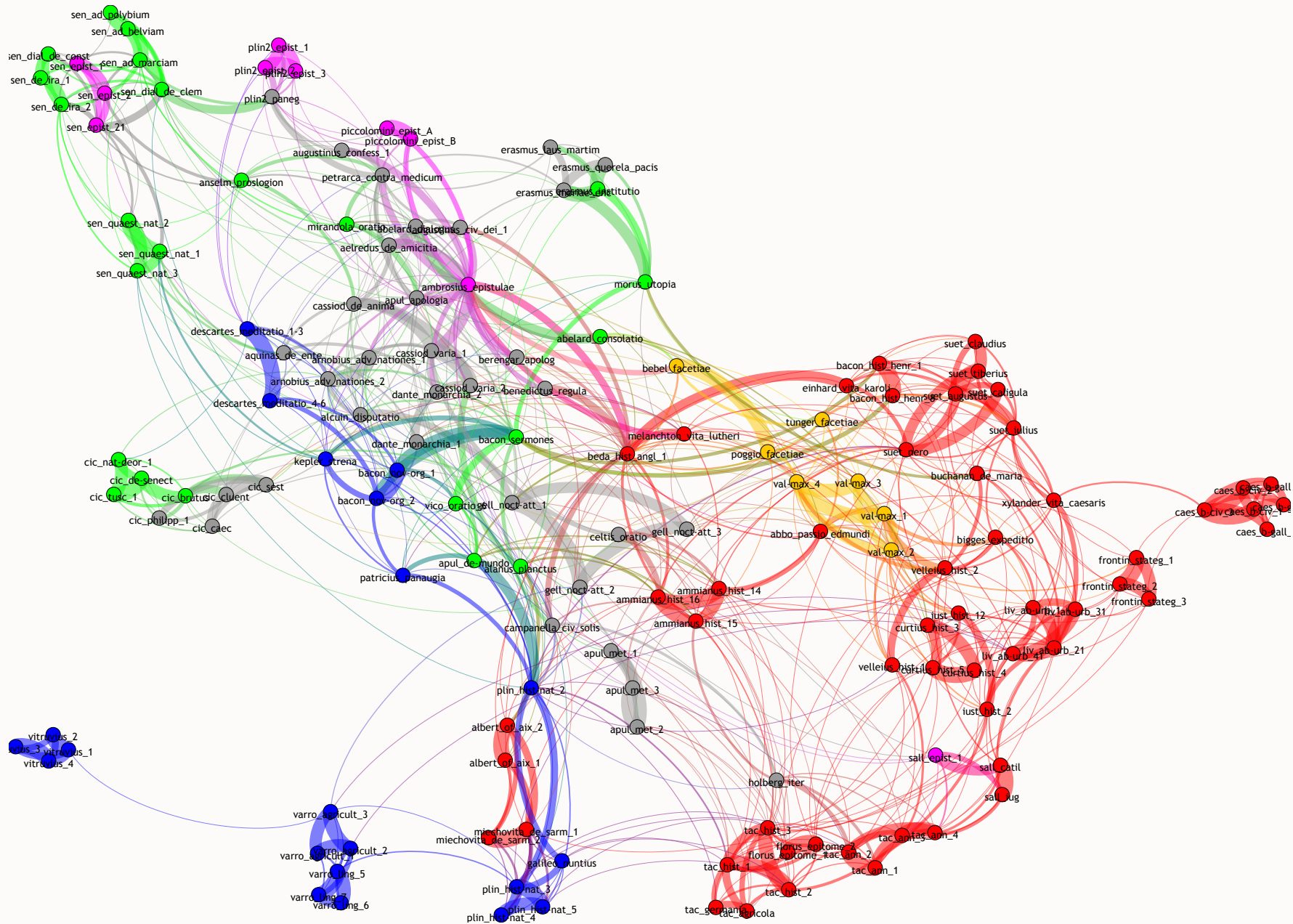
Latin literature at a glance

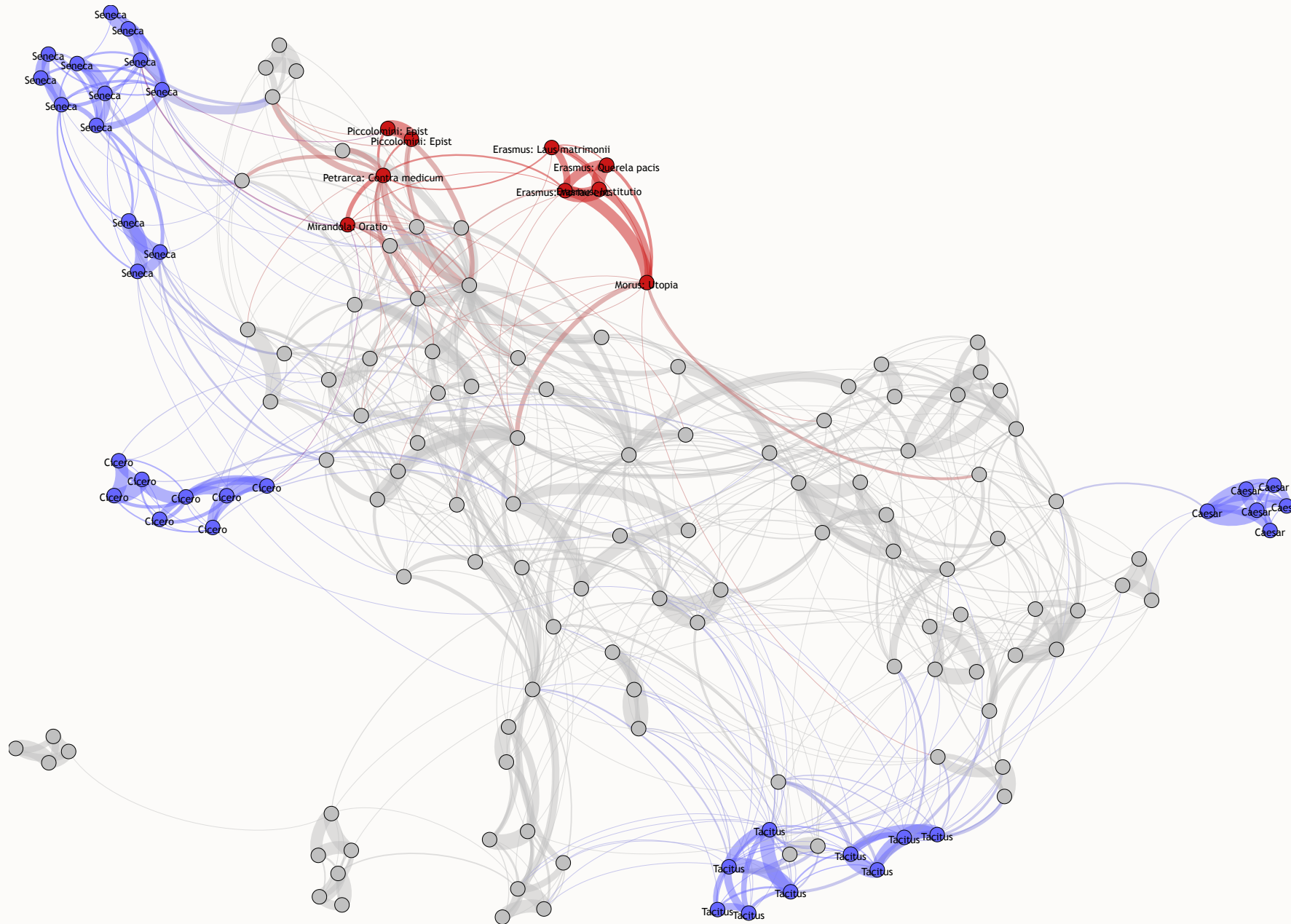


Traces of chronology?

- “Ciceronianus es, non Christianus!” (God accusing St Jerome)
- “Renovatio antiquitatis” (Renaissance humanists)
- hypothesis: noticeable traces of chronology

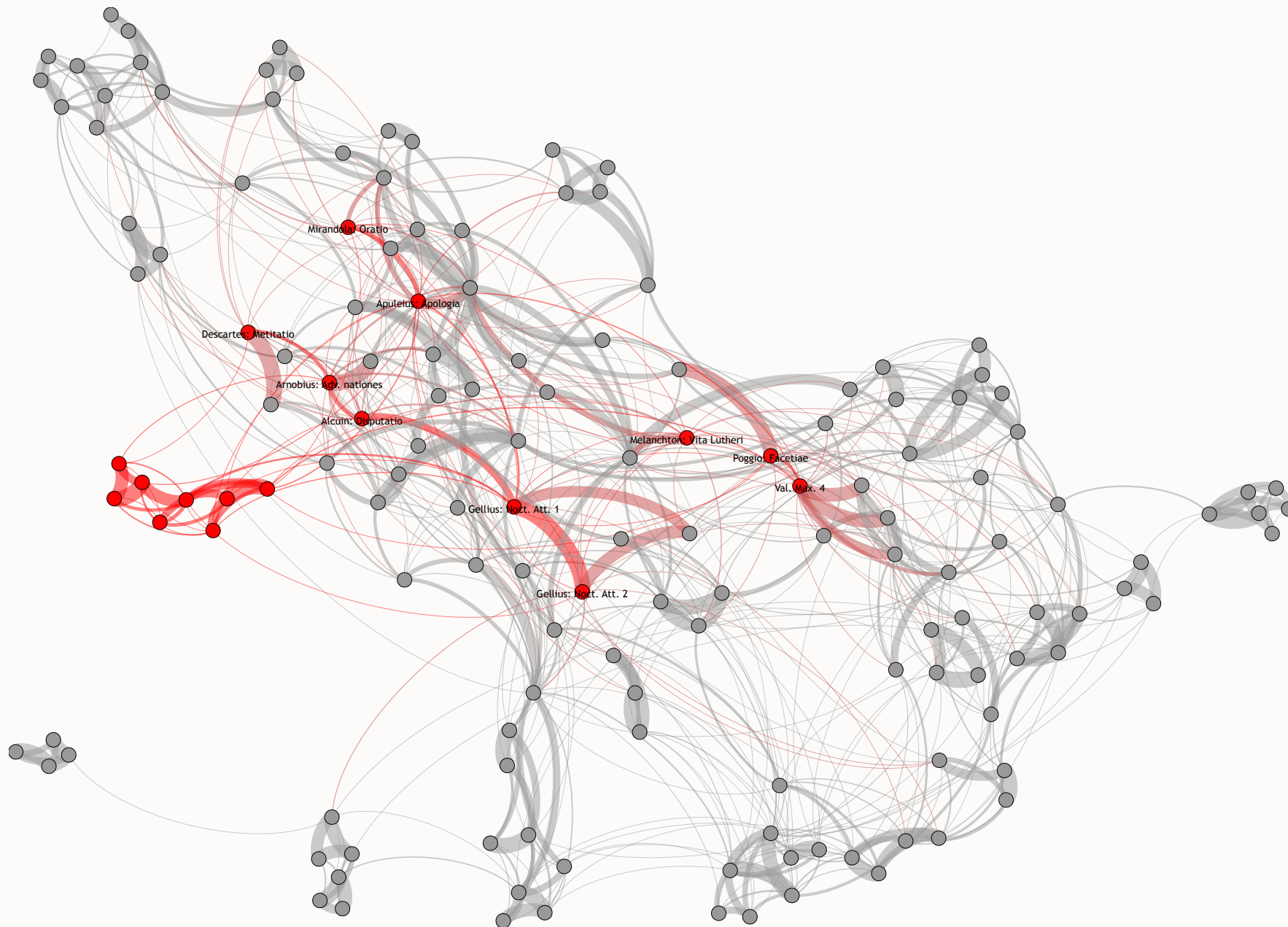






How influential was Cicero?

- Ciceronian Quarrel: the single most important debate of the Renaissance.
- Imitation of the admirable Ciceronian style.
- hypothesis: visible traces of Cicero in early modern Latin writings.



Conclusions

- No clear chronological pattern.
- Genre signal: predominant.
- Masters of style tend to keep outside the network.

Further research

- More texts!
- More genres!
- More “Attic” writers (e.g. Lipsius).